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CS 351 - Human-Computer Interaction project

Section: 6w9 - Group number: 1

Topic: Graduate Recruitment Application

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Table of contents:

Abstract:..... 3
Introduction: 3
The Project Problem / Idea: 4
The Project Purpose: 5
Analysis: 5
Users: 6
Tasks: 8
Requirements: 9
Interviews And Questionnaires: 11

Abstract:

Fresh graduates often struggle with online recruitment platforms that are mainly designed for experienced professionals. Complex profiles, unclear job filtering, and repeated data entry make the application process stressful and inefficient.

This project proposes a Graduate Job Portal interface tailored specifically for fresh graduates. The platform simplifies CV creation, improves job filtering for entry-level roles, and includes an AI career advisor to guide users toward suitable career paths. The goal is to create a smoother and more supportive transition from university to employment.

Introduction:

The transition from university to the job market is a crucial stage for fresh graduates. Although many online recruitment platforms are available, most are designed for experienced professionals, not beginners. This often makes the application process .confusing and discouraging for graduates

Current systems require detailed profiles and extensive work history, which many graduates do not have. In addition, job listings are not always clearly filtered for entry-.level roles, causing graduates to waste time reviewing unsuitable positions

This project proposes a recruitment platform interface designed specifically for fresh graduates. It focuses on simplicity, clear guidance, and highlighting academic projects, technical skills, and volunteer work. The aim is to provide a more structured and supportive job search experience

Project Problem and Idea:

When graduates move from university to the job market, they face a clear gap between their abilities and the design of current online recruitment systems. These platforms are mainly built for experienced professionals, which makes the graduate's first interaction stressful instead of supportive.

The problem is not only job availability, but how opportunities are presented. Platforms require long and detailed profiles, while graduates often lack enough work experience to complete these sections. This creates a feeling of inadequacy and turns registration into a psychological barrier.

The application process also lacks smooth flow. Repeated data entry, unclear steps, and poorly connected pages increase mental effort and reduce motivation. Instead of guiding users, the system confuses them.

Job search tools are another issue. Results often mix entry-level roles with positions that require years of experience, without clear filtering. Graduates waste time reviewing unsuitable jobs instead of applying for relevant ones. Searching becomes exhausting rather than productive.

These systems also overlook what graduates actually offer, such as academic projects, courses, technical skills, and volunteer work. Suitability is judged mainly by work experience, placing graduates at a disadvantage and affecting their confidence.

The issue lies in system design, not in graduates themselves. A recruitment platform built from a beginner's perspective is needed—one that is simple, provides clear guidance, reduces unnecessary steps, and highlights suitable entry-level opportunities. This would turn job searching into a structured and manageable path toward a first professional role.

Project Purpose:

The purpose of the Graduate Job Portal is to help and support newly graduated individuals in easily finding their dream jobs. It will freely allow them to upload and create a professional CV, filter the best jobs based on their skills and interests, and also include an AI career advisor that helps to discover the best career paths based on their skills, goals, and also suggest courses and skills they need to improve to get their dream job.

Analysis:

Current job portals such as LinkedIn and Glassdoor are helpful for finding jobs, but it might be complex for newly graduated individuals. Even if these applications provide a huge number of job opportunities, some people, especially newly gradated wouldn't make a real good use of them. So new application systems must focus on fresh graduates.

Users:

Primary Users: Fresh Graduates Seeking Employment

These users aim to secure entry-level job opportunities after graduation. They use the system to upload their CVs, search and filter job listings, and apply quickly to suitable positions. The portal supports them by simplifying the job application process and saving time and effort.

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Graduate Persona 1:

Name: Noura Muhammad

Age: 23 years

Background:

Noura is a recent Computer Science graduate who is actively searching for her first professional job. She applies to multiple positions weekly and prefers platforms that make the process simple and efficient.

Objectives:

- Find an entry-level job as quickly as possible
- Apply to multiple jobs with minimal effort
- Reuse her CV without re-entering information

Challenges:

- Time-consuming job application processes
- Job portals with complex navigation
- Job listings that do not match her skills

Secondary Users: Employers / Recruiters

These users use the system to post job vacancies, review graduate applications, and identify suitable candidates efficiently. The system helps them organize applications and reduce the time required for screening.

Employer Persona 2:

Name: Ahmad Al-Harbi

Age: 35 years

Background:

Ahmad is an HR recruiter working at a technology company. He is responsible for hiring fresh graduates and reviewing large numbers of CVs.

Objectives:

- Post job vacancies easily
- Identify suitable graduate candidates
- Review CVs efficiently

Challenges:

- Managing a high volume of applications
- Identifying qualified candidates quickly
- Poorly structured candidate information

Tasks:

• For fresh graduates:

1. They will register and create an account using basic personal information.
2. They will create or upload their CV through the platform.
3. They will search for entry-level job opportunities.
4. They will filter job listings based on skills, interests, and job type.
5. They will apply to multiple jobs quickly through the system.
6. They will receive notifications about job updates and application status.

• For employers / recruiters:

1. They will create an account and post job vacancies.
2. They will review and view graduates' CVs and profiles.
3. They will filter applicants based on qualifications and skills.
4. They will manage job postings and applications.
5. They will contact suitable candidates through the platform.

Requirements:

The following requirements have been categorized into Functional and Non-Functional requirements, ensuring all identified user needs from the research phase are addressed using standard HCI terminology.

Functional Requirements (FR):

A. Candidate Features (Primary Users)

1. User will be able to create a professional profile that highlights academic achievements, graduation projects, and university training (COOP).
2. The system will provide professional CV templates designed specifically for fresh graduates.
3. The system will analyze the user's profile and suggest relevant online courses to fill any detected skill gaps.
4. User will be able to filter job listings by major, GPA requirements, and "True Entry-Level" status.
5. The system will send automated smart notifications for jobs that exactly match the user's academic background.
6. User will be able to export their system profile as a professional PDF document.
7. The system will provide a "Career Directory" to explain job titles and their suitability for different majors.
8. The system will send push notifications for job matches.

B. Recruiter Features (Secondary Users)

1. The system will display a "Quick-View Card" for each candidate containing: Major, GPA, University, and Graduation Year.
2. User will be able to filter applicants using advanced academic filters.
3. The system will rank candidates automatically based on their compatibility with the job description.
4. User will be able to save selected candidates into a "Shortlist" folder for later review.
5. User will be able to add private ratings and internal notes on candidate profiles for team collaboration.
6. The system will allow recruiters to share candidate lists with other department managers.

Non-Functional Requirements (NFR)

A. Usability & User Experience (HCI Focus)

1. Efficiency: The system will allow users to complete a job application in 3 steps or less.
2. Learnability: A new user will be able to navigate the portal and set up a basic profile within 5 minutes.
3. Aesthetics: The interface will use a professional and clean color palette (e.g., Deep Blue and White) to enhance readability and trust.
4. Feedback: The system will show a real-time progress bar while the user is completing their profile.
5. Touch Targets: All buttons must have a minimum size of 44x44 pixels to ensure ease of tapping.
6. Haptic Feedback: The system will provide visual or haptic feedback (vibration) after successful applications.

B. Performance & Security

1. Response Time: All search results and AI-matching scores must be displayed within 2 seconds.
2. Privacy: The system will hide the candidate's personal contact information from unverified recruiters.
3. Availability: The portal shall be accessible 24/7 with a minimum uptime of 99.5%.
4. Security: The system will require multi-factor authentication for recruiter accounts to protect candidate data.

Interviews and Observation:

Interview Questions:

We conducted extensive semi-structured interviews with three different user groups to cover the entire recruitment ecosystem.

A. Upcoming Graduates (Seniors) Questions:

1. Since you are about to graduate, have you started exploring job portals? What was your first impression?
2. Since you lack work experience, how do you plan to convince companies of your abilities? (GPA, Graduation projects, or personal skills?)
3. Do you prefer the website to provide ready-made templates specifically designed for university students, or do you prefer designing your own CV and uploading it as a PDF?
4. Do you believe that your university projects (like the graduation project) deserve to be displayed prominently and separately rather than just a single line in a CV?
5. If the website informs you that your dream job requires a skill you don't have, would you prefer it to suggest training courses immediately?
6. Do you find difficulty understanding the current job titles? Do you wish for a "simplified guide" that explains which jobs exactly fit your major?
7. If the website provided a "Smart Alerts" feature based on your (GPA, University, and Major), would you rely on it as your primary platform?
8. As a student, do you prefer using a mobile phone for quick applications, or a laptop for careful data formatting?

B. Fresh Graduates Questions:

1. Tell me about your journey searching for your first job. Which platforms did you use, and how was the overall experience?
2. When you open a job portal (like LinkedIn), what is the first thing you do? How do you search for entry-level roles?
3. How much time do you usually spend completing your profile? Did you find any difficulty in filling out the required data?
4. Did you face a situation where you felt the website was "frustrating" because you lack previous experience (e.g., mandatory fields with no data)?
5. Do you feel that the skills you learned at university (projects, field training) appear clearly on current platforms?
6. When search results appear, do you find many jobs requiring "3-5 years of experience" even though you searched for entry-level? How does this affect your motivation?
7. If you could replace the traditional CV with something more expressive (like a portfolio), would you use it? Why?
8. How much do you need a feature that tells you the missing skills in your profile to become eligible for a specific job?

9. In a job application page, what information do you wish was clearer to feel confident before clicking "Apply"?
10. Do you prefer registering via long steps on one page, or dividing the process into small, simple stages?

C. Recruiter (Secondary User) Questions:

1. What are the top criteria you use for filtering fresh graduates (GPA, University, Major)?
2. How do you manage and organize a large number of graduate applications?
3. Would a private "Rating and Notes" system help you in collaborating with your team?
4. Do you value graduation projects when evaluating a candidate with no work experience?

Interviews Results Summary:

The interviews with 6 seniors, 3 graduates, and 1 recruiter yielded the following insights:

- Academic Identity: There is a unanimous agreement among seniors (Participant 1, 4, 5, 6) that graduation projects reflect their hard work and should be showcased as "Practical Evidence" of their skills.
- The Experience Paradox: Graduates (Participant 1, 3) highlighted the "Frustration" of mandatory experience fields. They feel current platforms ignore their potential because they focus solely on work history.
- Skill Gap & Guidance: Users (Seniors & Graduates) feel a strong need for direction. They confirmed that they would rely on a platform that suggests courses and explains job titles.
- Device Preference: Most users (Participant 1, 4, 5) emphasized their preference for a mobile-first approach due to the Immediacy of job alerts and the ability to apply on the go. They expressed that current websites are too complex for the mobile screens, which justifies the need for our dedicated Mobile Application.
- Recruiter's Perspective: Efficiency is key. The recruiter emphasized that "Candidate Cards" with quick data (GPA/Major) would save hours of screening.

Observation Findings:

A naturalistic observation was conducted by monitoring fresh graduates during their job search activities. The following behaviors were recorded:

- Time Consumption & Frustration: Graduates spent an average of 2-4 hours creating a single profile. Observation showed signs of boredom and distraction when faced with long, single-page forms.
- The Experience Barrier: Users were observed repeatedly skipping "Mandatory Work Experience" fields or filling them with irrelevant data (like "None" or "Fresh Graduate") just to bypass the system, which indicates a poor UX design for this segment.
- Search Inefficiency: Users spent significant time filtering out jobs that required 3+ years of experience. Even when using filters, irrelevant "Senior" roles appeared, leading to visible signs of decreased motivation and "Search Fatigue."
- Data Discrepancy: It was observed that university skills (COOP/Graduation Projects) often get merged or lost within the "Education" section on traditional sites, making it hard for the user to prove their technical competence.
- Participants were observed struggling with small clickable elements on mobile browsers.
- Users often missed important details due to poor mobile scaling on current job sites.